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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/270,733	07/03/2023	Cédric CARON	21335US01	5648

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EXAMINER	
HELLNER, MARK	

ART UNIT	PAPER NUMBER
3648	

NOTIFICATION DATE	DELIVERY MODE
02/19/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/270,733	Applicant(s) CARON et al.	
	Examiner MARK HELLNER	Art Unit 3648	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☐ Responsive to communication(s) filed on _____.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 7/3/2023; 1/23/2024.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Priority

Receipt is acknowledged of certified copies of papers required by 37 CFR 1.55.

Information Disclosure Statement

The information disclosure statements filed 7/3/2023 and 1/23/2024 have been considered by the examiner.

Drawings

The drawings filed 7/3/2023 are approved by the examiner.

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

Claims 1-3, 6-9, 12, 13 and 16-20 are rejected under 35 U.S.C. 102(a)(2) as being anticipated by Meynants (United States Patent Application Publication No. 2021/0223371).

With respect to claim 1, Meynants discloses: Time-of-flight demodulation circuitry **[taught by the abstract]** configured to: determine a light event pattern with an event-based light detection element of a plurality of event-based light detection **elements [paragraph [0075] states,** *"…The control unit CTRL controls operation of the image sensor IS, the driver logic DRV and the output stages OUT. The control involves timing of the pixels in the array, e.g. in an imaging mode IM…";* **detection of an image meets an event]**; and determine, for a demodulation element of a plurality of demodulation elements, a timing for a demodulation

signal to be applied to the demodulation element based on the light event pattern, wherein the demodulation element is associated with the event-based light detection element **[paragraph [0075] states,** "*…and in a time-of-flight mode TM according to an imaging timing sequence and according to a time-of-flight, TOF, timing sequence, respectively. The control unit CTRL determines timing, exposure and bias of the individual pixels in the array…"***]**.

Claim 7 is anticipated by the subject matter of Meynants, as applied to claim 1.

With respect to claims 2 and 8, [paragraph [0103] states, *"…The imaging mode IM includes a global shutter timing sequence to operate a pixel as shown in FIG. 3. The global shutter timing sequence can be applied to all pixels in the image sensor or only to part of pixels, e.g. in a defined region of interest, ROI…"*; **part of the pixels defining a region of interest meets a subset; thus, anticipating claims 2 and 8.**

With regard to claims 3 and 9, a region of interest, such target objects in figure 8, create a light event pattern in the sensed image; thus, anticipating claims 3 and 9.

The 2D image produced during the imaging mode meets representing ambient light, thus anticipating claim 6 and 12.

With respect to claim 13, Meynants discloses: A time-of-flight imaging apparatus **[taught by the abstract]** comprising: a plurality of event-based light detection elements each configured to detect a light event, based on which at least one light event pattern is determined **[taught by the image sensor (IS) in figure one operating in the imaging mode; paragraph [0075] states,** *"…The control unit CTRL controls operation of the image sensor IS, the driver logic DRV and the output stages OUT. The control involves timing of the pixels in the array, e.g. in an imaging mode IM and in a time-of-flight mode TM according to an imaging timing sequence and according to a time-of-flight, TOF, timing sequence, respectively. The control unit CTRL determines timing, exposure and bias of the individual pixels in the array…"* **]**; a plurality of demodulation elements, wherein each demodulation element is associated with at least one

event-based imaging element **[taught by paragraphs [0078] to [0080]]**; a light source configured to emit modulated light **[paragraph [0026] states,** *"…In at least one embodiment an emitter is configured to emit a plurality of light pulses in response to respective trigger pulses of an emitter control signal. In the time-of-flight mode, the control unit generates the emitter control signal. The phases are defined with respect to the trigger pulses of the control signal…"* **]**; and control circuitry **[the control unit taught by paragraph [0075]]** configured to: determine the at least one light event pattern [an image produced by the imaging mode meets a light event pattern]; control the light source to emit modulated light based on the determined at least one light event pattern **[the pulse signal used in the TOF mode meets modulated light (pulse modulation)]**; and control a subset of the plurality of demodulation elements to apply a demodulation signal, if the at least one light event pattern is detected with a subset of associated event-based light detection elements [paragraph [0103] states, *"…The imaging mode IM includes a global shutter timing sequence to operate a pixel as shown in FIG. 3. The global shutter timing sequence can be applied to all pixels in the image sensor or only to part of pixels, e.g. in a defined region of interest, ROI…"*; **part of the pixels defining a region of interest meets a subset]**.

Claim 19 is anticipated by the subject matter of Meynants, as applied to claim 13.

Figure 1 shows the sensor elements for imaging and the circuits for demodulating TOF data integrated together, thus anticipating claim 16.

The 2D image produced during the imaging mode meets representing ambient light, thus anticipating claim 17.

Paragraph [0076] teaches a control resetting the pixels to control the start of an exposure period, thus anticipating deactivation, as set forth by claims 18 and 20.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 4, 5, 10 and 11 are rejected under 35 U.S.C. 103 as being unpatentable over Meynants (United States Patent Application Publication No. 2021/0223371).

With regard to claims 4, 5, 10 and 11, paragraph [0076] of Meynants states, *"…In more detail, the control unit CTRL comprises means, such as a dedicated logic or a microcontroller, for controlling exposure of pixels, timing sequences and bias of individual pixels or groups of pixels in the array. The control unit CTRL can control resetting of the pixels to control the start of an exposure period (e.g. including operation of transfer gates and reset via a control signal), operation of transfer gates to transfer charges to a charge storage node and/or floating diffusion fd, operation of switches by means of a control signal to control read-out from a pixel…"*; thus, implicitly suggesting a series of measurement frames.

Therefore, it would have been obvious for a person of ordinary skill to have had a reasonable expectation to have have configured the device of Meynants to have met the limitation of claims 4, 5, 10 and 11, when doing measurements over a series of exposure periods, as inferred by paragraph [0076].

Claims 14 and 15 are rejected under 35 U.S.C. 103 as being unpatentable over Meynants (United States Patent Application Publication No. 2021/0223371) in view of Ou et al (CN 111598072 A).

Claim 14 differs from Meynants by reciting a first imaging portion including the plurality of event-based light detection elements; and a second imaging portion including the plurality of demodulation elements.

Figure 1 of Ou et al teaches that it was known in combined range/imaging systems to have used a TOF system (111) and imaging system (112) in a side-by-side arrangement next to the transmission element (110).

` Therefore, it would have been obvious for a person of ordinary skill in the art to have had a reasonable expectation of success in modifying the hybrid detection array disclose by Meynants to use a side-by-side arrangement, as shown by Ou et al, because the Ou et al arrangement was an alternative structure producing the same expected result.

Claim 15 is met by the combination of Meynants and Ou et la, as applied to claim 14, because figure 1 of Ou et al shows the TOF system (111) between the transmission element (110) and imaging system (112).

Any inquiry concerning this communication should be directed to MARK HELLNER at telephone number (571)272-6981.

Examiner interviews are available via a variety of formats. See MPEP § 713.01. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

/MARK HELLNER/
Primary Examiner, Art Unit 3648